# Agatha Ulina Silalahi

Jakarta, Indonesia | silalahiagatha15@gmail.com | +6281375278931 | LinkedIn | GitHub | Portfolio | Hugging Face | Tableau

## Summary

Data Scientist and AI/ML engineer with a Master of Mathematics (Data Science) from Universitas Indonesia, currently doing NLP research on central bank communication at Bank Indonesia Institute. Builds credit risk models and LLM applications for financial services and measures them before release: temporal validation, calibrated baselines and release gates for models; hand-checked golden sets, guardrails and cost limits for LLM apps. Works across the stack from SQL and data pipelines to model serving, CI and Docker, and every project metric below can be re-run from a public repository.

## Education

**Universitas Indonesia**
Master of Mathematics, Data Science specialization (GPA: 3.61) August 2023 – Oct 2025
*Relevant Coursework:* Computational Data Science, Advanced Machine Learning, Applied Deep Learning, Advanced Statistical Analysis, Advanced Data Exploration and Visualization, Advanced Bioinformatics, Advanced Genomic Data Science.

**Pacmann Academy** - Non Degree Program 2023 – 2025
Job Preparation Program Basic: AI/ML Engineer
*Relevant Coursework:* Python Programming & Software Engineering, Probability & Statistical Testing, Linear Algebra, Machine Learning Algorithms, APIs & Deployment, Recommender Systems, Neural Networks, Machine Learning Product Management, Natural Language Processing.

**Universitas Negeri Medan**
Bachelor of Mathematics Education, FMIPA (GPA: 3.53) 2017 – 2021

## Work Experiences

**Bank Indonesia Institute (BINS)** | Data Scientist Research Assistant Mar 2025 – Present
- Build NLP pipelines that score Bank Indonesia press releases and related news for clarity, sentiment and coverage, for research on central bank communication.
- Combine lexicon-based scoring, machine learning classifiers and manual annotation to label sentiment in financial news.
- Use topic modelling and time series analysis to track inflation and exchange-rate themes, as input to an early warning indicator.

**PT Bank Rakyat Indonesia (Persero) Tbk** | Artificial Intelligence Research Intern Aug 2024 – Jan 2025
- Built a prototype LLM resume parser: an internal LLM prompted with a fixed JSON schema at temperature 0 extracts name, contact, education, skills and work history from CV text, served through a Flask web app.
- Turned unstructured CV text into structured records that HR could filter and compare during screening.
- Public version of the prototype with synthetic data: GitHub.

**SMA Global Mandiri Cibubur** | Mathematics Teacher Jun 2022 – Jun 2023
- Taught mathematics, prepared annual and semester programs, syllabi and lesson plans, and coached students in academic and non-academic activities.

**SMPK Penabur Gading Serpong** | Bilingual Mathematics Substitute Teacher Jan 2022 – Apr 2022
- Managed the class and taught mathematics bilingually (English and Indonesian).

**Private Teacher** | Mathematics Tutor 2020 – Present

## Project

- **NLP/LLM Engineering** | Project · Live Demo | **FinSight IDX: RAG Question Answering over Annual Reports** — Python, Claude API, BM25, ChromaDB, Tesseract OCR, Gradio, Docker
  - RAG app over the 747-page BCA 2023 annual report that cites the page of every figure, evaluated on a golden set of 58 questions checked by hand in the PDF (recall@k, MRR, answer, citation and refusal accuracy).
  - Audited my own index (duplicate chunks, wrong page numbers, skipped text) and fixed it with regression tests; indexing each table row as its own chunk raised page recall@5 from 0.30 to 0.58, and BM25 with 10 chunks reached 0.70.
  - Answers are correct for 79% of questions (65% before), cite the right page 91% of the time and refuse unanswerable questions, at about one US cent per question; rate limits, a daily cost ceiling, telemetry and 220 tests in CI.
- **Machine Learning** | Project | **Credit Default Early Warning System with Explainability** — Python, XGBoost, SHAP, scikit-learn
  - Default model on 307,511 Home Credit applications (8.1% default rate): XGBoost ROC-AUC 0.763 and PR-AUC 0.251, against 0.745 and 0.225 for an imputed, scaled logistic regression baseline.
  - The top 20% of scores captures 52% of defaults; probabilities are calibrated so they can feed expected-loss estimates.
  - Removed gender from the features and measured the cost in accuracy; SHAP shows two external bureau scores carry 29% of the model's attribution. Built a PSI drift detector and tested it on simulated drift.
- **Credit Risk Modelling** | Project · Dashboard | **CreditLens: Credit Scoring Research with a Release Gate** — PostgreSQL, dbt, XGBoost, FastAPI, Streamlit

  - Warehouse of 2,260,668 LendingClub loans in PostgreSQL and dbt, with a temporal train, validation and test split fixed before modelling.
  - Compared a constant baseline, logistic regression and XGBoost on the same cohort. The best model reached average precision 0.220 on the frozen test, below the 0.25 gate set in advance, so I held it back from release.
  - The FastAPI scoring service refuses unreleased models; a public Streamlit dashboard shows only aggregates; 158 automated tests.
- **Data Engineering** | Project | **End-to-End Data Pipeline for Credit Risk ML** — Python, PostgreSQL, XGBoost, Docker, Kubernetes, GitHub Actions
  - Five-stage ETL for 150,000 borrowers with identical metrics on PostgreSQL and SQLite; XGBoost ROC-AUC 0.866 and PR-AUC 0.394 against a 0.857 / 0.360 logistic baseline.
  - Removed World Bank macro features after proving they were constant (the dataset has no dates). Docker, a Kubernetes Job on local minikube, 11 tests and CI.
- **Data Analytics** | Project · Tableau · Looker Studio | **NusaCommerce E-Commerce Analytics Dashboard** — PostgreSQL, SQL, Python, Tableau Public, Looker Studio
  - SQL-first analysis of 20,848 Shopee orders; RFM on 409 buyer locations, where 88 locations generate 71% of revenue. Tableau and Looker Studio dashboards refreshed in October 2026.
- **Sentiment Analysis** | Project | Fine-Tuning a Pretrained BERT Model for Text Classification
- **Data Science Computation** | Project | Prediction of Medical Insurance Cost with an XGBoost Model
- **Data Science Statistics** | Project | Comparison of stress causes in developed and developing countries using clustering and multinomial logistic regression on the COVID Stress Global Survey dataset
- **Data Analyst Bootcamp DQLab** | Project | Time series forecasting of Superstore sales with an ARIMA model

## Skill and Training

**Skill:**

- **LLMs & Retrieval:** Claude API, prompt and output-schema design, retrieval-augmented generation (BM25, embeddings, ChromaDB, reciprocal rank fusion), PDF extraction and OCR, golden-set evaluation (recall@k, MRR, answer and citation accuracy).
- **Credit Risk & Evaluation:** ROC-AUC, PR-AUC, KS, calibration, temporal validation, bootstrap confidence intervals, PSI drift monitoring, release gates, removing protected attributes.
- **NLP:** Sentiment analysis, topic modelling, BERT fine-tuning, Indonesian financial text.
- **Programming:** Python (pandas, NumPy, scikit-learn, XGBoost, LightGBM, SHAP, SQLAlchemy) and R.
- **Data Engineering:** SQL on PostgreSQL (CTEs, window functions), dbt, ETL pipelines, REST API ingestion, export to Google BigQuery.
- **Engineering:** Git and GitHub Actions, pytest, Docker, FastAPI, Gradio, Streamlit, Hugging Face Spaces, Kubernetes on local minikube, rate limiting and cost control.
- **Data Visualization:** Tableau Public, Looker Studio, matplotlib, seaborn, ggplot2.
- **Statistical Analysis:** Hypothesis testing, regression analysis, clustering.

**Training:**

**AI/ML Engineering | Pacmann Job Preparation Program (JPP) Basic**

- Curriculum covering Python programming, probability and statistics, linear algebra, machine learning algorithms, machine learning APIs and deployment, and recommender systems, preparing for AI and machine learning roles.

**Bootcamp Data Analyst with SQL & Python in Google Platform Batch 8 | DQLab**

- Covered SQL and Python fundamentals, data visualization and storytelling with Looker Studio, and data science with Python; graduated with a time series forecasting project (ARIMA on the Superstore dataset).

## Certificate

**Fundamental of Probability & Random Variables and Statistical Test** | Pacmann Certificate
**Bootcamp Data Analyst with SQL & Python in Google Platform Batch 8** | DQLab Certificate